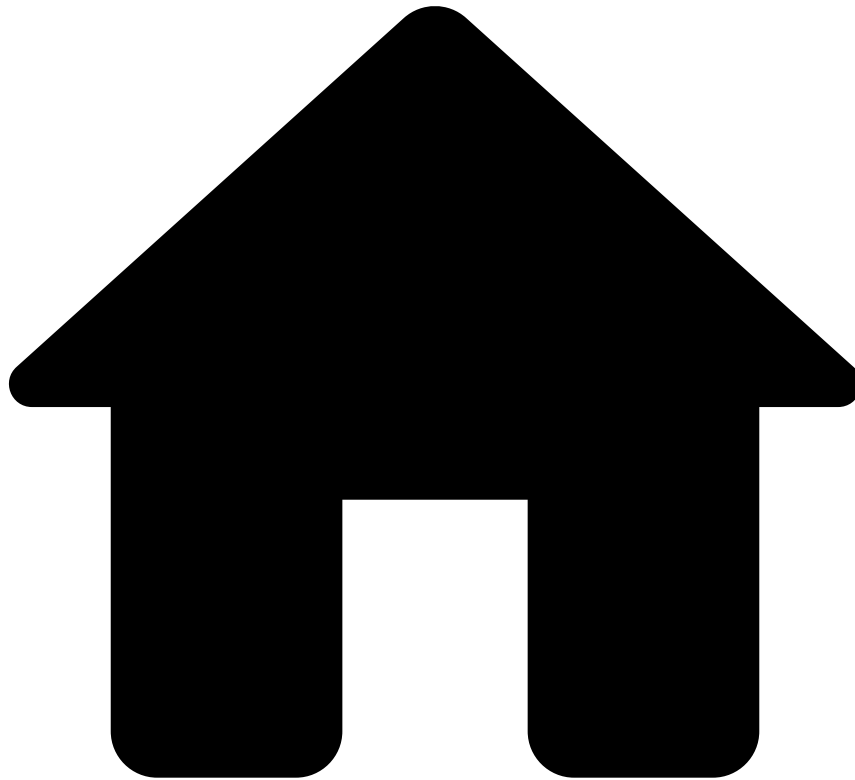


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- [Feature Engineering](#)
- Creating Derived Features

On this page

Creating Derived Features

Was this page helpful? 

This page describes how to edit and run an existing plan that creates a new data source with new features.

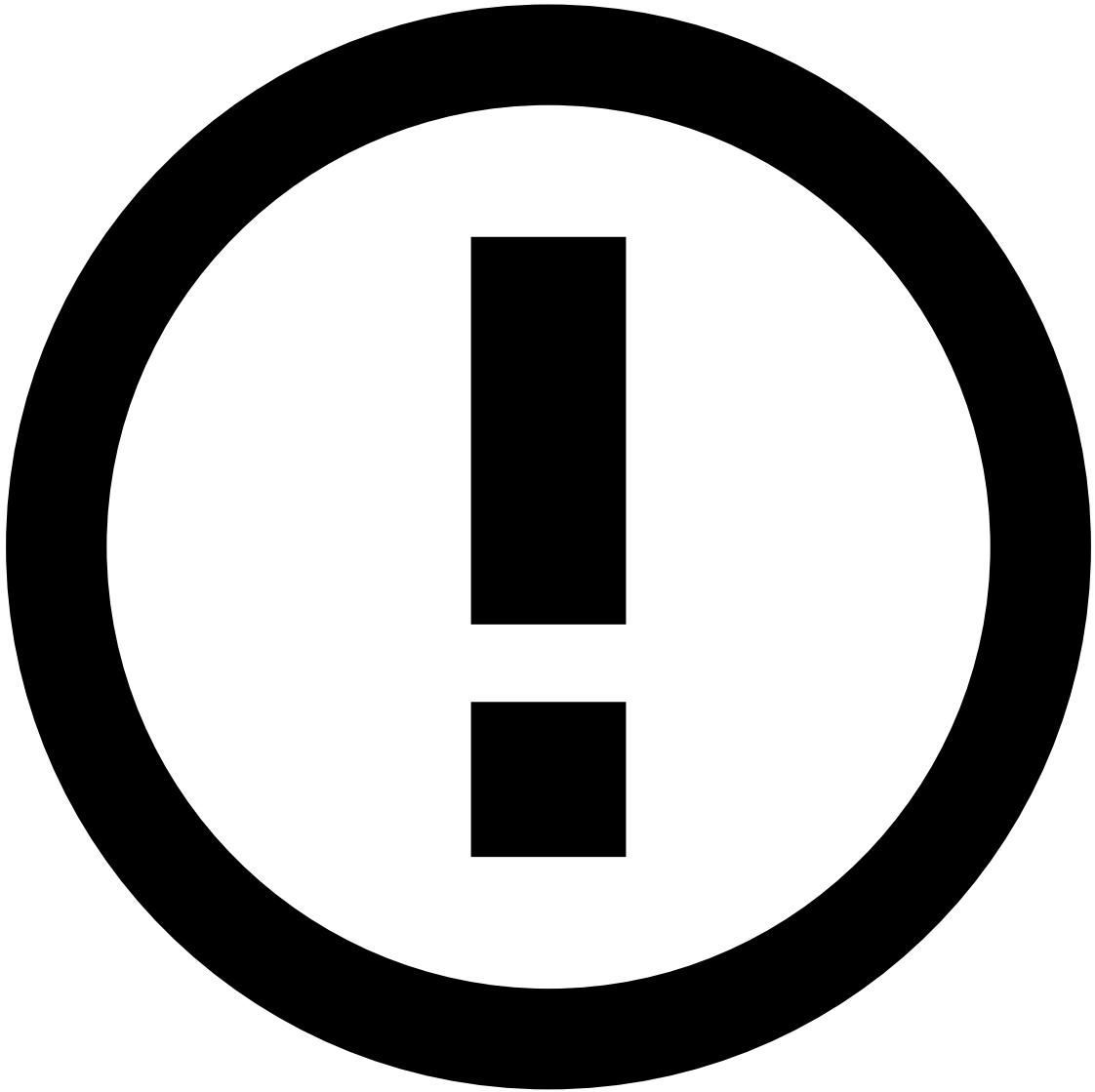
In the example shown, you can see how to add a new field that is calculated based on a single transaction. If you want to create a feature that is based on many aggregated transactions, you must use [metrics](#), i.e. historical profiles – refer to the page on [creating metrics](#).

Adding the features

To add new features, execute the following actions:

1. In the **Data Science** tab, open the features plan.

2. Drag a **Map** operator to the plan if there is none yet.



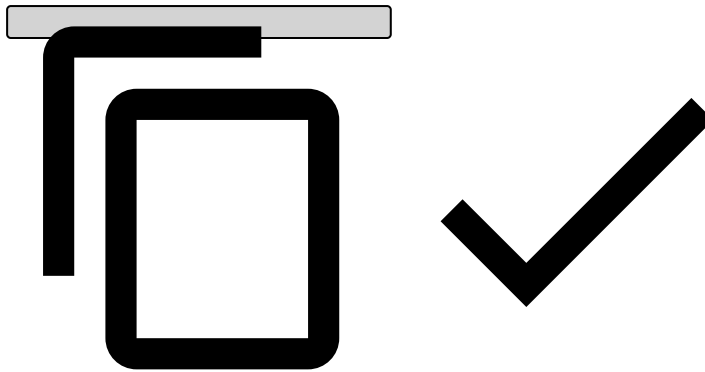
Clipboard

To save time, you can use the clipboard to copy-paste operators in plans. This allows you to add operators without the need of specifying all their fields each time you create a new plan.

3. Edit the map as follows:

1. Select the **Fields to Add** tab.
2. In the **Output Field Name** column, specify the name of the new feature.
3. In the **Expression** column, write the [Pulse Query Language expression](#) that calculates the output value.
For example, the following expression calculates whether the credit card was present at the transaction:

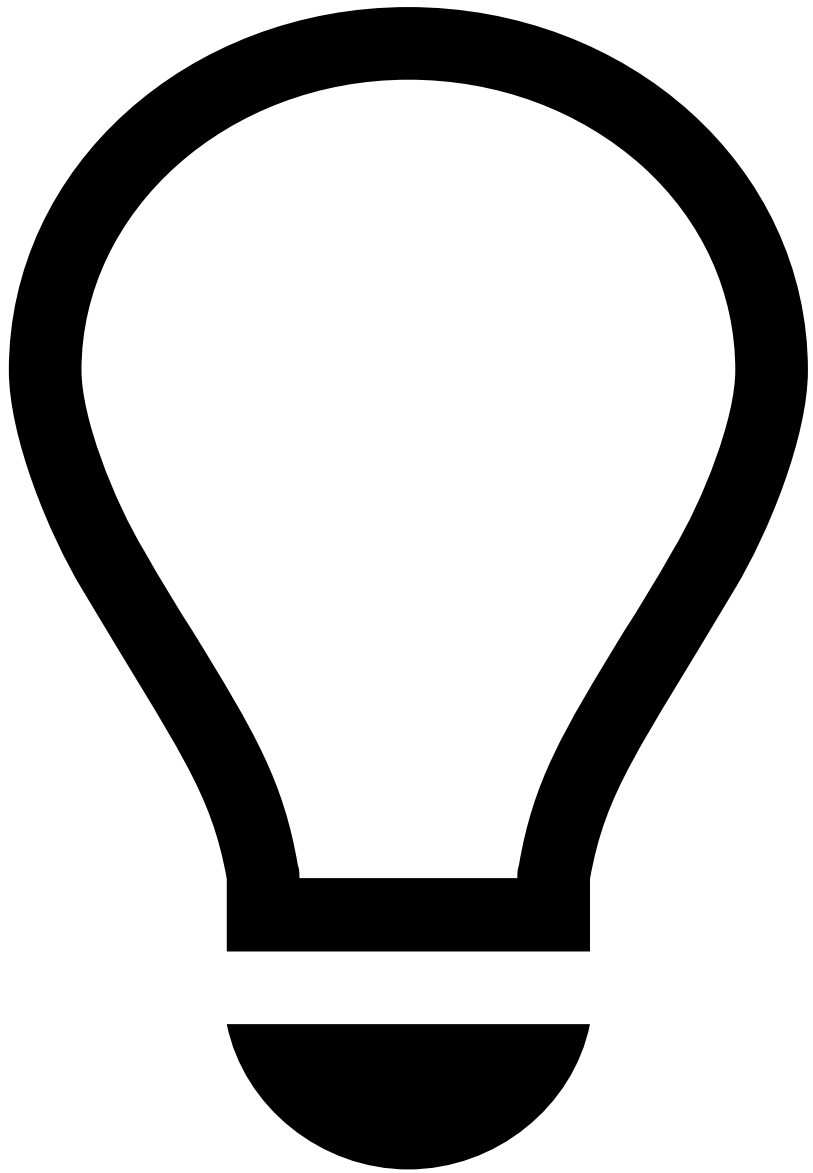
```
if (POSEntryMode == 'Swiped' or POSEntryMode == 'Keyed' or POSEntryMode == 'Chip')  
then 'true' else 'false'
```



4. Define the **Type** of the output field. In the example of calculating whether the credit card was present at the transaction, the type of the output field is *Categorical*, with two possible values: *true* and *false*.
5. Confirm that you want to add the field to the output schema, using the add button.
6. Add all the features to the map operator by following the previous steps.
7. **Save** the map operator.

Creating the data source with the features

1. To create a new data source which includes the new field(s), execute the following actions:
 1. With the plan still open, use the button to execute the plan.
 1. In the **Data Source** tab, select the data source that will be the input for the plan. For example *Cleaned Transactions*, or *Model Features v1*. If the plan has more than one channel, select the input data source for each channel.
 2. In the **Data Sinks** tab, configure the output of the plan:
 - **Name** for the output data source. For example, *Rule Features v1*. **Type**: *Parquet* (default) or *CSV*, depending on what type of output file you want to generate.



Parquet format is recommended

Using the Parquet file format for data files is highly recommended. Besides allowing a direct encoding of data types (which automates the mapping of field types when importing datasets), it also improves performance and disk usage: compared to CSV, Parquet files occupy less space and take less time to load.

- **Directory Path:** the path and file name where Pulse will store the output file, containing the data of the new data source. Example: `file:///datasets/rule-features-v1.parquet` . Note that the format varies according to the operating system.

3. Use the run button to execute the plan. It generates a new data source in Pulse, named *Rule Features v1*. Pulse also saves an output file with the transformed data to your filesystem.

2. Check the new data source to confirm that its schema includes new fields.